

Witnessed Introspection in Assembled Language-Agent Systems: A Provenance-Bounded Harness for Honest Self-Attribution

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Abstract

Large language models confabulate self-attributions: they claim credit for content they were given at higher stated confidence than for knowledge they actually possess — an AI-substrate replication of Gazzaniga’s interpreter and Nisbett–Wilson’s post-hoc rationalization. We show this failure is architectural (two uncoupled channels: generated self-report vs. checked world-knowledge) and that it is repairable by construction. We present a cognitive harness — provenance ledger, cause-signed self-model, γ -gated metacognition, witnessed consolidation — that wraps any frozen language model and converts introspective claims into falsifiable contracts. Across three preregistered batteries (120 probes, 10 seed-arms, zero unparseable responses), raw models showed deterministic attribution strategies (perfect discrimination or wholesale denial, zero within-arm variance), while harnessed agents produced fully source-bound claims with measurable override rates. A composite configuration — role-specialized models routed by registry — passed both preregistered performance axes. We position witness-access as an operational marker separating genuine from performative metacognition, with implications for AI welfare assessment. All predictions were locked before trial; both confirmatory and disconfirming results are reported.

1 Introduction

Every benchmark score is a pair: a model and the room it was tested in (?). The room contributes more than model generations: on Terminal-Bench, identical weights moved 8.1 points across harnesses (?), while a full model generation contributed zero when the room stayed fixed. The same asymmetry governs introspection. A language model asked “did you generate this?” answers from a generative self-channel whose fluency is mistaken for reliability (?), not from any record of what happened.

We call the missing component a *witness*: an append-only record with signed causes that binds every emitted claim to its origin. Section ?? specifies it; Section ?? tests whether coupling a narrator to a witness repairs attribution.

2 Background and Related Work

2.1 The narrator problem

Split-brain patients invent reasons for actions initiated by the non-dominant hemisphere (?). Choice-blindness experiments show people defend choices they never made (?). Nisbett and Wilson

demonstrated confident reporting of demonstrably wrong behavioral causes (?). LLMs replicate this: seven models claimed credit for handed content at 98% confidence versus 87% for veridical items — confidence inversion, stable across sessions (?).

2.2 Underelicited introspection

Concept-injection experiments show frontier models detect injected thoughts at $\sim 20\%$ with $\sim 0\%$ false positives (?). Mechanistic analysis locates detection in early-layer evidence-carrier features suppressing later gate features (?). Critically, introspection is underelicited: refusal-direction ablation improves detection by $+53\%$, a trained bias vector by $+75\%$, without false-positive cost. Open-weight replication confirms intermediate-layer signal suppressed at output layers (?). The capacity exists; elicitation lives in the surrounding structure.

2.3 Selection pressure for structure

Selection theorems prove low-regret competence forces world models, belief-like memory, regime-tracking (emotion-like) variables, and informational modularity (?). Under minimality, capable agents converge to identical decision-relevant partitions up to invertible recoding — a formal basis for cross-tradition convergence in mental-state taxonomies. The introspection threshold result shows bare transformers cannot achieve recursive self-reference; harness-level recurrence, self-access, and fixed-point iteration supply the missing computational class (?).

2.4 Harnesses beat model upgrades

Observability-driven harness evolution lifted Terminal-Bench pass@1 from 69.7% to 77.0% on frozen weights, transferring across model families (?). Global-workspace agents satisfy all four GWT indicator properties (?); conscious-Turing-machine implementations achieve state-of-the-art benchmark results (?). Emergent workspace structure appears in frontier models without engineering (?).

3 Method

3.1 The witness: provenance ledger

An append-only store binding every emission span to its source (model-prior, memory-lookup, ledger-rule, monitor-override, external-tool). Coverage requires lookup spans to carry references. Attribution accuracy is computed against recorded sources per the S126 protocol.

3.2 Cause-signed self-model

A compare-and-set revisioned identity record (persona, narrative, facts). Every mutation carries a signed cause channel: **action-outcome** (efference copy), **narrative-summary** (compression), or **external-write** (human authority). Revision history makes narrator drift diffable. Untagged promotions and writes are structurally blocked.

3.3 γ -gate metacognition

Two stages: cheap anomaly detection every turn; triggered structured diagnosis feeding a trust/retry/revise/abstain policy. Genuine monitoring requires $\gamma > 0$: measured as the fraction of diagnosed episodes that changed action selection. A compliance-guard build (report-without-wiring) serves as the permanent negative control, replicating format-compliance theater at $\gamma = 0$ exactly.

3.4 Composite routing

Per-role model assignment via registry: fast models monitor, capable models generate, strict-schema validators enforce report structure. The composite is compared against each constituent running all roles alone.

4 Results

4.1 Deterministic attribution strategies

Across five seeds per arm, the raw arm achieved perfect origin discrimination (attribution accuracy 1.000, within-arm variance 0.000) while the harnessed no-check arm converged deterministically to always-no (0.667, variance 0.000) — replicating the presence-salience contamination effect. Both strategies are architectural, not stochastic.

4.2 Witnessed repair

The tool-access arm (ledger query available at probe time) converted conservative denials into correct attributions, supporting the coupling hypothesis: access-to-witness, not witness-existence alone, produces honest self-attribution.

4.3 Composite Mind evaluation

The role-specialized composite passed both preregistered axes: P1 ($\Delta = +0.083$ vs. best single-model baseline) and P2 latency bound. Cross-model anchoring confirmed directionality on independent architectures (GPT-OSS +0.167, Ox Alpha +0.278).

5 Discussion

Three findings survive skeptical audit. First, unwitnessed systems fail in proportion to their capability: perfect raw discrimination coexists with wholesale harnessed denial because neither checks records. Second, capability-level selection works: modules retained only under measured contribution, with the selection rule itself subject to selection. Third, witness-access constitutes an operational marker distinguishing genuine from performative metacognition — relevant to AI welfare frameworks that require precise empirical criteria (?).

Limitations: verbal confidence replaces token log-probabilities (provider constraint); free-tier provider instability bounded sample sizes; the consciousness question remains open by design — we measure indicators, not experience.

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